

# Reading the v5.7 figure

13 September 2026 | Prompt A-prime angular model and existing HPS pair spectra

At fixed decay geometry, raising the beam energy shifts the accepted mass window upward. The figure makes that kinematic effect visible and places the recorded spectra beneath it. **The calculated curves are conditional angular fractions, not calibrated HPS signal efficiencies.**

## The 15 mrad scale and the upper edge

For an on-axis parent with energy  $E = x E(\text{beam})$ , equal-energy massless daughters have  $m = E \sin(\theta)$ . Here  $\theta$  is one track's angle to the beam; the pair opening angle is  $2\theta$ . At  $x = 1$ , the reference values are:

| Run  | Beam energy (GeV) | 15 mrad scale (MeV) | 70 mrad model edge (MeV) |
|------|-------------------|---------------------|--------------------------|
| 2015 | 1.056             | 15.84               | 73.86                    |
| 2016 | 2.300             | 34.50               | 160.87                   |
| 2021 | 3.740             | 56.10               | 261.59                   |

**A 15 mrad vertical gap alone imposes no upper mass cutoff.** The 70 mrad polar cap is an illustrative assumption motivated by a historical design requirement [4]. It is not a verified outer boundary of the complete HPS apparatus. The first row shows a vertical slice with angles enlarged for clarity, not a detector engineering drawing.

The lower scales are conditional too:  $x = 0.8$  moves them to 12.67, 27.60 and 44.88 MeV. A real parent has an energy and direction distribution. The seventh tracking layer and positron hodoscope used in 2021 also changed the detector response [1]. The common angular window isolates the energy dependence; it does not reproduce those changes. Neglecting electron mass shifts the tabulated  $x = 1$  symmetric edges by less than 0.04 MeV.

## What is integrated in the middle row

The parent decays promptly along  $+z$ . Its transverse-vector rest-frame distribution is  $(3/8)(1 + c^2)$ , with  $c = \cos(\theta^*)$  and uniform azimuth. Both boosted daughters must be forward and clear the vertical cut  $|\text{atan}(p_y/p_z)| \geq 0.015$ . The colored curves add  $\text{atan}(p_T/p_z) \leq 0.070$ . Grey dotted: gap only,  $x = 1$ . Solid: both cuts,  $x = 1$ . Dashed: both cuts,  $x = 0.8$ . The azimuth is integrated analytically and  $c$  numerically.

The denominator is all decays in this specified angular model. No production-spectrum weighting, polarization mixture, material, field transport, finite sensors, hit selection, trigger, reconstruction or displaced-decay acceptance is included. A physical efficiency curve requires matched simulated signal denominators and run-specific selections; a complete three-run set was not established here.

# Data, sources and validation

## Reading the bottom row

The curves are raw selected pairs per MeV in 1 MeV bins. They have no exposure or luminosity rescaling. They show the mass content of background-dominated samples, not A-prime production rates. Different selections and exposures prevent an energy-only interpretation of their heights.

| Sample    | Histogram                | Displayed pairs |
|-----------|--------------------------|-----------------|
| 2015 full | invariant_mass           | 21,442,838      |
| 2016 full | h_Minv_General_Final_1   | 73,218,251      |
| 2021 10%  | preselection/h_invM_8000 | 141,305,897     |

The inherited display crops are 15, 30 and 36 MeV; the figure ends at 300 MeV. The 2015 input ends at 150 MeV, so its grey region above 150 denotes unavailable bins. These crops and endpoints are not detector acceptance boundaries. The native files, exact bin values and source SHA-256 hashes are saved with the study. Missing/cropped CSV bins are marked explicitly.

Historical HPSTR radiative-MC acceptance products were inspected but not substituted for signal efficiency: they use SIMP control selections and SLIC-level truth denominators, with unverified equivalence to these data samples. No matched 2015 counterpart was established.

## Primary references

[1] N. Baltzell et al., The Heavy Photon Search Experiment (2022). Table I: beam energies; detector section: upgrades. [Source](#).

[2] P. H. Adrian et al., original 2015 dark-photon search (2018), p. 2: 1.056 GeV. [Source](#).

[3] P. H. Adrian et al., 2016 prompt/displaced search (2023), pp. 2, 5: 2.3 GeV and 15 mrad active edge. [Source](#).

[4] M. Battaglieri et al., HPS Test Detector (2015): approximately 15-70 mrad historical design criterion. [Source](#).

Additional 2021 trigger/acceptance context: V. Kubarovsky, 15 November 2021, slide 4 ([presentation](#)). Its curves were not digitized or used as the model efficiency.

## Checks and reproducibility

Count-preserving rebinning, fraction bounds, mass endpoints and mass/energy scaling passed. An independent two-dimensional integration of boosted momenta agrees with the analytic angular calculation within 0.000105 absolute acceptance at the checked points. A separate physics review confirmed the model formulas. Rendered-page inspection and extracted-text checks accompany the numerical results. These checks validate the model implementation, not the omitted detector effects.

Reproduce with `scripts/make_acceptance.py` and `scripts/make_methods.py`. One numerical worker, BLAS/OMP threads capped at one; no toys or detector simulation. CPU use was coordinated with the existing v5.5 and v5.6 tasks. See `README.md`, `references.json`, `derived/` and `qa/`.